Marmara University – Faculty of Engineering – Department of Computer Engineering

Fall 2020 – CSE3055 Database Systems
Quiz #4

(14.01.2021.Thu | Duration: 45 min)

1) [55 pts] Consider the following table with the given assumptions.

Employee (Ssn, FirstName, LastName, Gender, Age, Salary, DepartmentID)

Assumptions:

- There is no index.
- There are total 8,000,000 rows.
- 250,000 rows fall within the 2,500 to 3,000 salary range.
- There are 100 rows for each page.
- 400 index rows fit on a non-clustered index leaf page.

What is the number of page accesses of the following queries for the following cases with the given assumptions? Consider each case separately.

a) [28 pts]
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Select Ssn From Employee Where Salary>2500 and Salary<3000 and DepartmentID=1;
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b) [27 pts]
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Create Clustered Index ixEmpSal on Employee (Salary);
      Select Ssn From Employee Where Salary>2500 and Salary<3000 and Gender='M';
```

2) [45 pts] Consider the following *Appointment* relation:

Appointment (DoctorID, AppointDate, AppointTime, DoctorName, PatientID, PatientName, SurgeryNo)

Primary Key = {DoctorID, AppointDate, AppointTime} and you have the following functional dependencies:

- **FD1**: {DoctorID, AppointDate, AppointTime} ⇨ {PatientID, PatientName}
- **FD2**: {DoctorID} ⇨ {DoctorName}
- **FD3**: {PatientID} ⇨ {PatientName, SurgeryNo}
- **FD4**: {DoctorID, AppointDate} ⇨ {SurgeryNo}
- **FD5**: {AppointDate, AppointTime, PatientID} ⇨ {DoctorID, DoctorName}

According to the functional dependencies given above, which one(s) of them violates corresponding NF? **Explain each of them.**

a) [15 pts] First Normal Form (1NF): ❑ None ❑ FD1 ❑ FD2 ❑ FD3 ❑ FD4 ❑ FD5

b) [15 pts] Second Normal Form (2NF): ❑ None ❑ FD1 ❑ FD2 ❑ FD3 ❑ FD4 ❑ FD5

c) [15 pts] Third Normal Form (3NF): ❑ None ❑ FD1 ❑ FD2 ❑ FD3 ❑ FD4 ❑ FD5

IMPORTANT NOTES:

1) Textbooks, slides and notes are closed.

2) Internet usage is not allowed, however computers and cell phones may be used for only viewing this exam.

3) This exam consists of 2 questions and 2 pages.

4) Please write your answers clearly and neatly to the appropriate place on blank A4 pages.

5) Show all your work.

6) Write your student ID, first name and last name on top of each solution page.

7) Write the following sentences on the first page of your document: "I hereby swear that the work done on this quiz is totally my own; and on my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for this quiz. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues." Write your full name and sign it.

8) Solve all the questions in your handwriting on blank A4 sheets.

9) Your solution pages should be in the order given in this document.

10) Scan all the solution pages to a single PDF file. Your scans should be clear, readable, small in size, cropped and fixed, portrait oriented, upright and pages ordered.

11) Name your scanned file as "StudentID.pdf".

12) Upload your PDF file to http://ues.marmara.edu.tr before deadline.

13) In case of any form of copying and cheating on solutions, all parts will get ZERO points. You should submit your own work. In case of any forms of cheating or copying, both giver and receiver are equally culpable and suffer equal penalties. All types of plagiarism will result in zero points from the quiz.

14) Do not send quiz submissions through e-mail or any other media. E-mail attachments or such will not be accepted as valid submissions.

15) You are responsible for making sure you are turning in the right file, and that it is not corrupted in anyway. We will not allow resubmissions if you turn in the wrong file, even if you can prove that you have not modified the file after the deadline.

16) No groups are allowed.

17) Grade evaluation may be done on selected parts of the quiz, so try to complete all parts of your quiz successfully.

18) No late submissions will be accepted.